

2025 年 FRM 一级 Practices Exam 3 题目+知识点解析

◀ ▶ Question 1 of 100

□ Revisi

Section Instructions ▲

1. An economist from a central bank is presenting at a banking conference on the characteristics of deposit insurance and the effect on the overall risk profile of the banking system. During the presentation, the economist mentions that using deposit insurance might increase moral hazard risk. Which of the following examples can the economist use to illustrate a moral hazard?

- A.** A bank increases amount of equity capital it is holding to cover potential losses calculated at 99.9%.
- B.** A bank collects large amounts of deposits and reinvests those funds into risky high-yield bonds instead of high-quality bonds with lower returns.
- C.** A bank pays risk-based deposit insurance premiums on all its capital regardless of the amount of interest paid to depositors.
- D.** A bank decreases the interest rates it pays to depositors to compensate for deposit insurance premiums.

◀ ▶ Question 2 of 100

□ Revisi

Section Instructions ▲

2. A risk analyst at a fixed-income investment firm has acquired a rating transition matrix issued by a rating agency to use in estimating credit risks in the firm's bond portfolio. The analyst plans to use it to forecast rating changes in the upcoming year (Year 1) and the following year (Year 2). Which of the following would the analyst be correct to note when using the rating transition matrix in estimating credit risk?

- A.** An investment-grade bond is less likely to have its rating remain unchanged over the course of Year 1 than a speculative-grade bond.
- B.** A bond that is upgraded in Year 1 would be more likely to be upgraded in Year 2 than is indicated by the corresponding probability in the transition matrix.
- C.** Since external ratings are through-the-cycle, probabilities predicted for Year 2 using the matrix should remain accurate if the economic forecast for the year is revised significantly downward.
- D.** An assumption that rating changes in consecutive years are independent events produces the most realistic estimates of multi-year rating transitions.

3. A junior mortgage analyst has is analyzing a USD 2 million mortgage pool consisting of four mortgages. The original principal amount of the pool was USD 2.3 million. Additional information about the mortgages is provided below:

	Current loan balance (USD)	Interest rate	Maturity at origination (months)	Current maturity (months)
Mortgage A	350,000	3.60%	360	340
Mortgage B	475,000	3.75%	360	310
Mortgage C	550,000	4.12%	360	270
Mortgage D	625,000	4.25%	360	250

Which of the following are the best estimates of the weighted-average coupon (WAC) and the weighted-average maturity (WAM) of the mortgage pool?

- A. WAC 3.93%; WAM 286 months
- B. WAC 3.93%; WAM 293 months
- C. WAC 3.98%; WAM 286 months
- D. WAC 3.98%; WAM 293 months

4. An investor is testing the hypothesis that the mean return generated by an equity fund is zero and collects return data on the fund over the past 250 trading days. The investor calculates the sample daily mean return and the sample standard deviation as 0.2% and 0.35%, respectively. In performing the hypothesis test, which of the following is the most appropriate construction of the alternative hypothesis for testing the true mean, m ?

- A. Construct a one-tailed test with $H_A: m > 0$
- B. Construct a one-tailed test with $H_A: m < 0.2$
- C. Construct a two-tailed test with $H_A: m \neq 0$
- D. Construct a two-tailed test with $H_A: m \geq 0.2$

5. A bank has both term loans and interest rate swaps outstanding with many of its medium-sized business clients. A risk manager at the bank is comparing the processes used to estimate different metrics that are inputs into the Basel credit risk capital calculation for the term loans and the swaps. According to the Basel Committee, which of the following metrics requires the addition of an add-on amount to the current estimate when calculating credit risk capital for interest rate swaps?

- A.** Probability of default
- B.** Effective duration
- C.** Exposure at default
- D.** Recovery rate

6. A junior trader at a UK-based commodity trading company is considering implementing a trading strategy using futures. The trader notes that the Brent crude oil futures for delivery next month are currently trading below the spot price. Assuming no transaction or delivery costs, which of the following strategies would be expected to generate a profit for the trader?

- A.** Short the futures, buy the Brent crude at spot, and deliver against the futures at settlement.
- B.** Short the futures and deliver against the futures at settlement.
- C.** Buy the futures, short Brent crude at spot, and take delivery against the futures at settlement.
- D.** Buy the futures and take delivery against the futures at settlement.

7. A financial advisor is explaining the basics of bond trading to a client. The advisor focuses on the connections between the credit spread of a bond and that bond's yield. Which of the following statements would the advisor be correct to make to the client?

- A.** When the credit spread on a bond decreases, the yield on that bond will increase.
- B.** As the maturity of a bond increases, the credit spread increases for lower-rated bonds and decreases for higher-rated bonds.
- C.** US corporate bonds pay a higher yield than US treasury bonds to compensate the buyer for taking more risk.
- D.** Higher-rated bonds pay a higher yield because they have a higher credit spread.

8. A junior trader on a derivatives trading desk is using historical data to run a linear regression model that regresses the price of an option against the underlying stock price. The trader conducts an analysis of the regression results to identify potential outliers and assess their impact on the estimated coefficients. Which of the following statements correctly describes the given approach for identifying outliers?

- A.** A residual plot compares the error values on the x-axis against the realization of the explanatory variables on the y-axis.
- B.** Residual plots can only detect outliers that are typically within 4 standard deviations of the model mean.
- C.** A Cook's distance value greater than 1 indicates that an observation has a large impact on the regression parameters.
- D.** Cook's distance measures the sensitivity of the fitted values in a regression when the two largest observations are dropped.

9. A risk analyst is interpreting the results derived from applying a GARCH (1,1) model for estimating the current trading day's price volatility of a stock. Selected inputs to the model are provided below:

- Previous trading day's return (r_{n-1}): -3%
- Previous variance rate (σ_{n-1}^2): 0.0009
- Long-run average variance rate (V_L): 0.0001

Assuming α , β , and γ are held constant, how did the price volatility estimate calculated using GARCH (1,1) change from the previous day's value to the current day's value?

- A.** The estimate increased due to the effect of the previous trading day's return.
- B.** The estimate decreased due to the effect of the previous trading day's return.
- C.** The estimate increased due to the effect of the long-run average variance rate.
- D.** The estimate decreased due to the effect of the long-run average variance rate.

10. A manager at an OTC options dealer is reviewing the calculations of an analyst that were used to implement a delta hedge for a long position in a call option. The manager discovers that at the time the hedge was established, the analyst had estimated the delta of the option position as 60,000, but it was really 70,000. If the underlying stock has increased in price since the implementation of the hedge, which of the following statements is correct?

- A.** The value of the option position has increased, but this increase is larger in magnitude than the decrease in value of the stock position used to establish the hedge.
- B.** The value of the option position has increased, but this increase is smaller in magnitude than the decrease in value of the stock position used to establish the hedge.
- C.** The value of the option position has decreased, but this decrease is larger in magnitude than the increase in value of the stock position used to establish the hedge.
- D.** The value of the option position has decreased, but this decrease is smaller in magnitude than the increase in value of the stock position used to establish the hedge.

11. An options trader is applying the Black-Scholes-Merton option pricing model to estimate the delta of a long 1-year European put option on a dividend-paying stock. Relevant data is provided below:

- Annual continuously compounded risk-free rate of interest: 2.0%
- Annual dividend yield on the stock: 3.5%
- $N(d_1)$: 0.5411
- $N(d_2)$: 0.4287

What is the delta of the put option?

- A.** -0.5517
- B.** -0.5411
- C.** -0.4589
- D.** -0.4431

12. An internal auditor at a commercial bank is reviewing the risk management department's credit risk calculations relating to a small business loan in the bank's portfolio. The auditor finds the following information in the calculation:

- Expected loss rate: 4.62%
- Probability of default: 6.79%

What recovery rate was assumed by the risk management department in calculating the expected loss?

- A.** 32%
- B.** 47%
- C.** 68%
- D.** 89%

13. A hedge fund manager believes that energy prices are likely to fall due to lower demand for air travel and container shipping. To reflect this belief, the hedge fund manager enters into a short position on 20,000 shares of stock GRY, a large diversified oil and gas producer, at a price of CAD 60.00 per share. Stock GRY pays a regular quarterly dividend of CAD 1.50 per share and its previous dividend was paid out a week before the manager established the short position.

Seven months later, the manager closes out the short position when stock GRY is trading at CAD 49.00 per share. Assuming no changes in stock GRY's dividend policy occurred during the past seven months, the manager incurred no borrowing costs, and dividends are paid out on schedule, what is the net profit for the hedge fund manager on the trade?

- A.** CAD 160,000
- B.** CAD 190,000
- C.** CAD 220,000
- D.** CAD 280,000

14. The CIO of a pension fund is assessing several actively managed equity funds to add to the pension fund's portfolio. As part of this assessment, the CIO examines the suitability of various measures to evaluate the performance of outside managers and considers how each measure is calculated. Which of the following statements is correct regarding the given performance measure?

- A.** The Sharpe performance index calculates the ratio of the return of a portfolio to the return of the benchmark against which the portfolio is compared.
- B.** The tracking error calculates the standard deviation of a portfolio's active return.
- C.** The Sortino ratio calculates the ratio of the return of a portfolio to the standard deviation of returns above the portfolio's target rate of return.
- D.** The information ratio calculates the difference between the return of a portfolio and the return of the benchmark against which the portfolio is compared.

15. A derivatives trader at an investment bank has initiated an interest rate swap contract today with an institutional client. The current value of the trader's position in the contract is near zero. The trader considers using either DV01 or effective duration to measure the interest rate sensitivity of the position and assess by how much its value could decline if the term structure of spot rates moves unfavorably. What would be the most appropriate measure for the analyst to use to quantify the interest rate risk of the contract, and why?

- A.** DV01, because effective duration expresses interest-rate sensitivity as a currency amount rather than as a percentage
- B.** DV01, because the effective duration calculated for this position may not be meaningful in these circumstances
- C.** Effective duration, because DV01 cannot be calculated for a position in a derivative contract
- D.** Effective duration, because DV01 is only accurate when estimating the effect of a large change in interest rates

16. A risk analyst at a bank is assessing the suitability of applying modern portfolio theory (MPT) in portfolio analysis. The analyst starts by evaluating the assumptions underlying MPT. Which of the following is correct for the analyst to identify as a concern about the application of MPT in practice?

- A.** MPT assumes asset returns are asymmetric with fat tails, but empirical evidence shows that return distributions are typically normally distributed.
- B.** MPT assumes asset returns are normally distributed, but empirical evidence shows that return distributions are typically asymmetric with fat tails.
- C.** MPT assumes asset returns are normally distributed, but empirical evidence shows that return distributions are typically asymmetric with thin tails.
- D.** MPT assumes asset returns have negative skewness, but empirical evidence shows that return distributions typically exhibit positive skewness and fat tails.

17. A risk manager on the corporate bond desk at an investment company is explaining the important metrics used to measure and manage the risks of bonds to a group of newly hired analysts. The manager describes the concepts of recovery rates and default rates. Which of the following statements is correct for the manager to make?

- A.** Recovery rates follow lognormal distribution with a single mode, shaped by differences in the seniority levels among bonds from the same issuer.
- B.** The issuer default rate relates the number of bonds that a specific issuer has defaulted on to the total number of bonds outstanding from this issuer.
- C.** Recovery rates used in financial analysis are typically calculated as the post-liquidation amount of principal retained by issuers.
- D.** The dollar default rate uses par values to measure the value of bonds that have defaulted as a percentage of the total value of outstanding bonds.

18. A risk consultant is advising a US-based bank on the Dodd-Frank Act stress tests (DFAST), an annual stress test exercise conducted by the US Federal Reserve (the Fed) for mid-size US banks. The consultant points out that the Fed requires the banks to utilize three supervisor-devised macroeconomic scenarios during the DFAST stress test. Which of the following describes the conditions that one of the supervisor-devised scenarios is designed to simulate?

- A.** A set of economic variables reflecting the forecast by the Fed's chief economist
- B.** Economic conditions that correspond to the average projections from a survey of economic forecasters
- C.** The worst market conditions experienced in a 10-day period during the previous year
- D.** An economic expansion with an associated rapid increase in demand for real estate investments and fixed-income instruments

19. A small bank sets the interest rates it charges on loans to its business clients at a level equal to the sum of the loans' expected loss, the bank's funding cost, and a margin for the bank's profit and operating expenses. A risk analyst at the bank compiles the following information related to the loan portfolio:

Average recovery rate of loans	45%
Margin for profit and operating expenses	2.10%
Average funding cost	4.85%
Average interest rate on loans	8.18%

Given this information, what is the estimated average probability of default for the loans in the portfolio?

- A.** 0.68%
- B.** 1.23%
- C.** 2.24%
- D.** 2.73%

20. A junior risk analyst at an asset management firm is monitoring the performance of a recently launched mutual fund against a benchmark index. The analyst uses the last 36 months of excess returns data to construct a confidence interval that can be used to test the one-sided hypothesis that the average excess monthly return of the mutual fund is greater than 0%. Information about the hypothesis test is given below:

- A 5% significance level is used.
- Excess monthly returns are assumed to follow a normal distribution.
- The standard deviation of excess returns over the previous 36 months is 0.05.
- The average excess monthly return over the previous 36 months was 1.2%.

Which of the following provides the correct confidence interval to be used as the decision criterion for the hypothesis test?

- A.** $(-\infty, -0.17\%]$
- B.** $[-0.17\%, \infty)$
- C.** $[-0.17\%, 2.57\%]$
- D.** $[1.20\%, 2.57\%]$

- 21.** A junior analyst at a bank is examining the trading activities of different derivatives market participants. The analyst focuses on understanding the motivations and goals of hedgers, speculators, and arbitrageurs. Which of the following is a correct example of a hedging activity?
- A.** A soybean farmer purchases a 6-month out-of-the-money call option on soybeans.
 - B.** A trader, who believes that a commodity price is mean-reverting over time and is currently undervalued, enters a long position in the commodity.
 - C.** A trader enters a transaction where the forward price of the asset is greater than the price at which it could be purchased today by borrowing the full purchase price.
 - D.** A US-based manufacturing company that is scheduled to make a large payment in euros in 1 year purchases a 1-year forward contract for the same euro amount.
- 22.** An enterprise risk manager at a large US-based bank is reviewing the bank's stress testing and scenario analysis approaches to recommend potential improvements. The manager prepares a presentation describing some benefits of stress testing and scenario analysis as well as the regulatory and historical context of these two approaches. Which of the following statements is correct?
- A.** Sensitivity testing involves substituting several key variables with alternative variables to explore risk model results under stressful conditions.
 - B.** Scenario analysis is used to model the effects that abnormal events and events with no historical data could have on the bank's performance.
 - C.** In the Comprehensive Capital Analysis and Review (CCAR) stress test, banks with assets above USD 50 billion are limited to the use of a single severely adverse scenario developed by supervisors.
 - D.** Scenario analysis that included the assessment of the impact of cumulative exposures across multiple business lines were widely used across the banking industry even prior to the 2007-2009 financial crisis.

23. An analyst at Bank FLR is calculating the cash flows on a forward rate agreement (FRA) that the bank entered into 1 year ago. The bank is a fixed rate payer under the FRA and the terms of the FRA state that interest accrues for the 3-month period beginning 1 year after initiation. The analyst collects the following information about the FRA and interest rates in the market:

- Principal amount: EUR 10,000,000
- Fixed interest rate: 2.25%
- Floating interest rate 9 months after initiation of the FRA: 2.05%
- Floating interest rate today: 1.80%
- Market forecast of floating interest rate 3 months from now: 1.90%

What is the amount of the cash flow required today to settle the FRA?

- A.** EUR 4,975
- B.** EUR 5,000
- C.** EUR 11,200
- D.** EUR 11,250

24. A risk manager at a foreign exchange dealer is assessing the relative level of real and nominal interest rates in the US and Canada. The exchange rate is currently at CAD 1.21 per USD 1, and the firm's economist forecasts an exchange rate of CAD 1.19 per USD 1 one year from now. The analyst notes that the annual risk-free rate of interest in the US is 0.85%, and the annual inflation rates for the US and Canada are 2.50% and 2.80% respectively. If purchasing power parity and covered interest parity both hold, what is the best estimate of the difference between the real interest rates in Canada and the US?

- A.** 0.30%
- B.** 1.36%
- C.** 1.69%
- D.** 1.91%

25. A performance analyst at an asset management firm is measuring the returns of individual positions in a fixed-income portfolio. Details regarding the performance of a specific bond over a recent 6-month period are as follows:

- Purchase price on July 1: BRL 1,062,500
- Coupon received on September 30: BRL 27,500
- Coupon received on December 31: BRL 27,500
- Closing price on December 31: BRL 1,048,200
- Total financing cost from July 1 to December 31: 3.5%
- Return on reinvested funds from September 30 to December 31: 3.0%

What was the net return earned on the bond over the period from July 1 through December 31, annualized using semi-annual compounding?

- A.** -2.26%
- B.** 0.82%
- C.** 3.91%
- D.** 7.82%

26. A senior investment advisor at a wealth management firm manages an investment portfolio consisting of two main assets, fund A and fund B, for a group of clients. The annual returns of these assets can be represented as random variables, denoted as A and B, respectively. The advisor wants to estimate future portfolio performance by calculating the expectation of a specific function that is based on the anticipated returns of both funds A and B. The function is given by:

$$g(A, B) = 0.6A + 0.4B$$

where the coefficients reflect the portfolio allocation of 60% to fund A and 40% to fund B.

In the analysis, the advisor assumes that the returns of both fund A and fund B can take on only two possible year-end values, and constructs the joint probability mass function (PMF) as follows:

		A	
B		8%	11%
	1%	0.15	0.20
	3%	0.40	0.25

What is the correct expectation, $E[g(A,B)]$, of the function?

- A.** 2.30%
- B.** 5.12%
- C.** 6.53%
- D.** 9.35%

27. A compensation analyst at a consulting firm is comparing defined benefit pension plans and defined contribution pension plans offered by a company to its employees. The analyst prepares a detailed analysis of the structures and characteristics of these pension plans. Which of the following is correct regarding defined contribution pension plans?

- A.** Funds contributed to the pension plan are kept in separate accounts for each employee.
- B.** An employee's risk of not meeting a minimum retirement benefit level is assumed first by the employer, then by government-sponsored entities.
- C.** The company determines how the funds will be invested so that the maturity of the plan's assets matches the maturity of its obligations.
- D.** The present value of pension obligations is estimated each year to determine the employer's contribution.

28. A risk manager at a bank is giving a presentation to a group of interns on lessons learned from financial crises. The manager focuses on the case of the savings and loan (S&L) crisis in the US during the 1980's. Which of the following is most appropriate for the risk manager to conclude as a lesson to be learned from this case?

- A.** Financial institutions face a disadvantage when issuing floating-rate residential mortgages that are funded by short-term liabilities.
- B.** Maintaining a negative interest rate spread between lending and borrowing rates is a profitable business model for financial institutions.
- C.** Financial institutions should not be bailed out if the sole funding source for the bailout is other, larger financial institutions.
- D.** Issuing mortgage-backed securities helps provide financial institutions with liquidity for their mortgage portfolios.

29. A risk manager is examining the impact of counterparty credit risk in the OTC derivative markets. The manager notes that OTC market participants have adopted several different techniques to mitigate this credit risk. Which of the following correctly characterizes one of these techniques?

- A.** Requiring the recalculation of the collateral value and posting of the collateral on a quarterly basis
- B.** Providing for netting agreements that offset customized bilateral OTC derivative exposures across multiple counterparties
- C.** Requiring a credit default swap to be embedded in a derivatives contract to protect both parties from a default of the other party
- D.** Creating special purpose vehicles (SPVs) that legally separate the credit risks of the assets held within the SPV from the credit risk of the company that sponsors the SPV

30. The board of directors of a mid-size bank has recommended that the bank adopt the Basel principles for data aggregation and risk reporting. The bank currently has three different business units for its retail banking, commercial banking, and credit card operations. It also maintains some internally developed models and is introducing artificial intelligence (AI) techniques in data management. In assessing the impact and best practice of complying with the Basel principles for data aggregation and risk reporting, which of the following statements is most appropriate?

- A.** Business units of the bank should adopt departmental applications and methodologies for data management that are customized for each business unit.
- B.** Complying with the Basel principles should reduce the bank's exposure to model risk as the quality of data input in a model will likely improve.
- C.** An exponential increase in the application of AI techniques in data management is not likely to impose additional challenges to the compliance with Basel principles.
- D.** The Basel principles recommend that the bank outsource its model validation function to an external third party in order to provide an independent review.

31. A credit risk manager at an insurance company is examining the credit ratings on sovereign bonds held in the company's investment portfolio. The manager researches the factors that rating agencies consider when determining sovereign credit ratings. Which of the following statements is correct regarding the process used by rating agencies to rate countries?

- A.** Rating agencies use a country's government debt-to-GDP ratio as the primary debt ratio for assessing the country's sovereign default risk.
- B.** Rating agencies use a country's total debt-to-GDP ratio as the primary debt ratio for assessing the country's sovereign default risk.
- C.** A country with a higher government debt-to-GDP ratio will always receive a lower rating than a country with a lower ratio.
- D.** A country with an economy concentrated in a few industries will tend to have a more stable tax base, which has a positive impact on its total debt-to-GDP ratio and its rating.

32. The CRO of a local bank is examining the latest internal bank-wide risk report. The CRO notes that there are several key risk indicators implying a decrease in the amount of the bank's current credit exposure. Which of the following observations is most likely an indication that the bank's credit risk is decreasing?

- A.** The number of lending customers has increased significantly while the average size of a loan in the bank's lending portfolio has decreased.
- B.** The bank's sources of funding have been shifting from longer-term funding sources to short-term repo financing.
- C.** The bank's total value of retail time deposits on its balance sheet has increased while the value of wholesale deposits has decreased by the same amount.
- D.** The number of employees at the bank's retail branches has decreased as customer usage of its digital banking platform has increased substantially.

33. Responding to recent market news that a large hedge fund is experiencing difficulties, the CEO of a bank requests information regarding the bank's exposure to that fund and other major counterparties. The CRO of the bank reviews the bank's credit exposures and notes that the bank has exposures to hedge fund counterparties across different lines of business, which are not aggregated. The CRO recommends implementing an ERM program to improve the bank's risk governance. Which of the following actions would be most appropriate for the CRO to suggest?

- A.** Manage different classes of enterprise-wide risk using transfer instruments specific to each class.
- B.** Designate the board of directors as the responsible party for making day-to-day risk management decisions.
- C.** Aggregate risk exposures across risk types but maintain hedging strategies within the business lines.
- D.** Design a compensation plan that rewards business activities that benefit the bank at the enterprise level.

34. A portfolio manager at a mutual fund is evaluating the end-of-year returns of the stocks in the portfolio and comparing them to those predicted by the Fama-French model. The manager considers stock GRI, which had a return of 7.8% over the previous year and had a beta to the market of 1.25. The factor betas for the stock and the returns of each factor are presented in the following table:

Factor	Factor beta	1-year factor return
HML factor	-0.45	2.50%
SMB factor	2.10	1.00%

The market had a return of 6.00% over the previous year and the risk-free rate of interest remained constant at 1.50%. What was the alpha of stock GRI for the previous year?

- A.** -2.55%
- B.** -2.18%
- C.** -0.30%
- D.** 1.20%

35. A risk manager at a commodity trading company wants to reduce the firm's risk exposure by selling 1,000 kilograms of commodity ST with a 1-year forward contract. Before entering into the position, the manager wants to estimate the fair forward price of commodity ST and gathers the following information:

- Spot price of commodity ST: JPY 5,201 per kilogram
- Annualized lease rate: 2.25%
- Present value of the annual storage cost for commodity ST: JPY 65 per kilogram
- Annually compounded risk-free interest rate: -0.35%

Assuming zero convenience yield, which of the following is the best estimate of the fair 1-year forward price of commodity ST?

- A.** JPY 5,005
- B.** JPY 5,132
- C.** JPY 5,366
- D.** JPY 5,403

36. A risk consultant is presenting to a group of bankers and investment managers about the importance of maintaining a strong and effective risk culture at a firm. The consultant begins by discussing some challenges to the establishment of a strong risk culture. Which of the following correctly describes a challenge to firms in ensuring a strong risk culture?

- A.** Risk culture typically develops at the local business unit level, making it harder for local teams to adopt a firm-wide risk culture.
- B.** Risk culture can only be assessed effectively during normal market conditions and not during times of stress due to market volatility.
- C.** Risk is typically expressed in a common manner throughout an organization, making it more difficult to identify concentrations of risk.
- D.** A strong risk culture requires that the board of directors performs day-to-day risk management activities, but boards typically take an advisory role.

37. The CRO of a commodity trading firm is reviewing the hedging strategies used by the traders at the firm. The CRO notes that the traders often use dynamic hedging strategies to manage the price fluctuations of their positions and that they apply a wide-ranging limit system in controlling the hedging positions. Which of the following conclusions would the CRO most likely reach when reviewing the hedging strategies of the traders?

- A.** Stop loss limits are typically used to limit realized losses.
- B.** Concentration limits are very effective in capturing correlation risks during stressed market conditions.
- C.** VaR models are not affected by classic model risks, which are typically associated with traditional mathematical models.
- D.** Sensitivity tests are limited to measuring the change in value of derivatives.

38. A finance intern at a wealth management firm constructs a linear regression model that regresses the return of a stock on the market index return. After running the regression, the intern finds that the estimated coefficient for the market index return is 0.569 and that the resulting R^2 measure is 0.781. Which of the following would the intern be correct to calculate as the correlation between the return of the stock and the market index return?

- A.** 0.610
- B.** 0.754
- C.** 0.781
- D.** 0.884

39. A risk analyst at a wealth management company is calculating the ES of a portfolio. The portfolio's returns are expected to follow a uniform distribution, with all returns between 18% and -16% being equally likely. What is the 97% ES of the portfolio?

- A.** -15.49%
- B.** -15.15%
- C.** -14.98%
- D.** -14.30%

40. A newly hired model risk analyst at a bank is validating a model that was recently developed for use in stress tests conducted by the bank. The analyst first consults a senior colleague regarding best practices for validating stress testing models. Which of the following statements would the analyst's colleague be correct to make?

- A.** Stress test models are typically back-tested against the bank's historical data to ensure that the model is specified correctly.
- B.** Each stress test should use a single model so that the relationship between the scenario variables and the output is clear and unambiguous.
- C.** Models should produce a range of values for the bank's loss in a stress scenario, rather than a point estimate.
- D.** Relationships between variables in a model used for stress testing should typically be represented using the averages of their correlations over one full economic cycle.

41. A quantitative risk manager at a hedge fund is using an autoregressive moving average (ARMA) process to model the default premium of corporate bond portfolios held by the fund and constructs several covariance-stationary ARMA models with varying lag lengths. The manager conducts a graphical analysis of the autocorrelation function (ACF) and partial autocorrelation function (PACF) plots to identify candidate models. In this analysis, the manager notes that there is a slow decay in the ACF plot, but the PACF plot indicates a sharp cutoff at p lags. In determining whether an autoregressive (AR) or a moving average (MA) component is needed in the model, which of the following would be correct for the manager to conclude?

- A.** The plots indicate that the model should include an AR component.
- B.** The plots indicate that the model should include an MA component.
- C.** The plots indicate that the model should include both an AR and an MA component with lag length p .
- D.** The plots indicate that the model should incorporate an MA component after undergoing differencing for lag p .

42. A risk manager is evaluating the impact of using the delta-normal model in calculating VaR for the following two positions:

- A long position in a 6-month call option on 10,000 barrels of West Texas Intermediate crude oil (WTI) with a strike price of USD 85 per barrel
- A long position in futures contracts to purchase 10,000 barrels of WTI in 6 months at a price of USD 85 per barrel

The current spot price of WTI is USD 84.50 per barrel. In assessing the 1-day 95% VaR of the two positions, which of the following is correct?

- A.** The VaR of the futures position is equal to the VaR of the call option position.
- B.** It would be inappropriate to assume that the expected value of the daily change in the WTI spot price is zero.
- C.** As the spot price of WTI decreases, the VaR of the call option will decrease in a linear fashion.
- D.** The delta-normal model assumes that the 1-day change in the value of the call option is normally distributed.

43. An investment manager at a large investment bank is developing a new MBS product to offer to the bank's clients. The manager asks an analyst to conduct a thorough analysis of the different types of MBS available in the market and the securitization process used to create these products. Which of the following is correct for the analyst to conclude about securitization after the analysis?

- A.** Securitization is a practice used exclusively by financial institutions to remove assets from their balance sheets.
- B.** The securitization process is designed to prevent moral hazard from arising in the loan underwriting process.
- C.** Securitized products and credit derivatives share the same structure of pooling assets and repackaging them into interest-bearing securities.
- D.** Securitization increases the liquidity of the market by permitting the distribution of risk among a broad range of market participants.

44. A credit manager at a US-based bank is preparing a presentation to a group of interns on the role of credit derivatives in the 2007-2009 financial crisis and on subsequent changes in the credit derivative markets. The manager describes some characteristics of credit derivatives and also discusses some regulatory changes unique to the credit derivative markets. Which of the following statements is correct regarding credit derivatives?

- A.** Credit derivatives can facilitate the transfer of credit risk without incurring significant funding liquidity risk.
- B.** Securitization is used primarily for repackaging corporate bank loans and is limited in its use for collateralized loan obligations.
- C.** The required retention of credit risk for originators of asset-backed portfolios was increased from 10% to 25% by the US Securities and Exchange Commission after the crisis.
- D.** Credit derivatives markets are effective in measuring default risk over longer time horizons but are ineffective in measuring default risk in real time.

45. A risk analyst at a large investment bank is classifying various loss events experienced by the bank into the categories of operational risk identified by the Basel Committee. The analyst studies the case of a trader at the bank who unintentionally increased the size of a position in the bank's account above its VaR limit, and the bank incurred a loss when unwinding the position. Under which of the following Basel operational risk categories would this event most likely fall?

- A.** Internal fraud
- B.** External fraud
- C.** Clients, products, and business practices
- D.** Execution, delivery, and process management

46. A risk analyst at a bank is assessing modern portfolio theory (MPT) and the validity of applying MPT in asset pricing. The risk analyst begins by evaluating the assumptions underlying MPT. Which of the following is correct for the analyst to identify as a concern when applying MPT in practice?

- A.** MPT assumes that investors ignore the skewness of a return distribution.
- B.** MPT assumes that return distributions are asymmetric.
- C.** MPT assumes that asset returns can be explained by a small number of systematic factors that affect all securities.
- D.** MPT assumes that by using diversification, investors can eliminate both systematic and idiosyncratic risks from their portfolios.

47. A researcher is running a regression to test the hypothesis that the forward interest rate is an unbiased predictor of the future spot rate. The researcher uses a large sample size but is concerned about the potential impact of heteroskedasticity on the estimates of the ordinary least squares parameters. If the regression residuals exhibit heteroskedasticity, which of the following observations is most likely for the researcher to make?

- A.** The coefficient estimates will be biased.
- B.** The standard error estimates for the slope coefficients are too small.
- C.** The explained sum of squares will be equal to 1.
- D.** The standard deviations of the residuals are too low.

48. A quantitative analyst is preparing a performance report of a corporate bond portfolio. For the previous 1-year period, the analyst finds that the return on the portfolio was 4.4% and decomposes this return as follows:

- Return attributable to the risk-free interest rate: 3.4%
- Return attributable to the credit spread: 2.0%
- Loss rate over the period: 1.0%

When observing the portfolio's returns over a 3-year period, the analyst notes that the difference between the portfolio's return attributable to the credit spread and its loss rate remained positive and varied from 0.7% to 1.3% each year. Which of the following would the analyst be correct to identify as the most likely explanation for the persistent positive difference between the credit spread and the loss rate?

- A.** Bond investors are being compensated for their exposure to systematic risk.
- B.** Credit spreads are specifically set by market makers to be greater than loss rates.
- C.** Corporate bonds have more liquidity than risk-free sovereign debt.
- D.** Credit spreads tend to be negatively correlated with interest rates.

49. A data science team at the market-making division of a global financial institution is working to improve the division's risk-adjusted returns on options trading using machine-learning methods. The team creates a reinforcement learning (RL) algorithm that lets the learning agent develop price setting strategies. The strategies are intended to maximize profit and minimize inventory risk by dynamically adjusting bid and ask prices in the market maker's electronic limit order book.

The RL algorithm defines the following market states and market actions:

- State 1: high transaction costs and high volatility
- State 2: low transaction costs and low volatility
- Action 1: take a market-neutral approach
- Action 2: take a directional approach

The team considers using either the Monte Carlo (MC) approach, formulated as

$$Q_{\text{new}}(S,A) \leftarrow Q_{\text{old}}(S,A) + \alpha[R - Q_{\text{old}}(S,A)]$$

or the temporal difference (TD) learning approach, to update estimates for the Q-values, called $Q(S,A)$ or the value of taking an action in a particular state called, used to decide on the optimal strategy.

The current Q-values for the 2-state, 2-action scenario are given in the table below:

	State 1 (S1)	State 2 (S2)
Action 1 (A1)	0.7	0.8
Action 2 (A2)	0.5	0.9

In the next trial of the RL algorithm, the team notes that under an α parameter of 0.05, the MC approach samples a run taking A2 in S2 and estimates the total subsequent reward to be 1.2. Furthermore, the next decision on the trial is when the learning agent is on S1, with the jump between S2 and S1 generating a reward of 0.3. S1 is where the TD approach is initialized.

In using the two approaches, which of the following provides the correct update on the $Q(2,2)$ -value?

- A.** Using the MC approach, $Q(2,2)$ would be updated to 0.915.
- B.** Using the MC approach, $Q(2,2)$ would be updated to 1.200.
- C.** Using the TD approach, $Q(2,2)$ would be updated to 0.895.
- D.** Using the TD approach, $Q(2,2)$ would be updated to 1.000.

50. A risk analyst at a high-yield bond fund is calculating the conditional probability of default (PD) for a CCC-rated bond. The analyst has the following cumulative PD data from a rating agency:

Year	1	2	3
Cumulative PD (%)	24.78	34.51	39.86

What is the conditional probability that the bond defaults during year 3?

- A.** 5.35%
- B.** 8.17%
- C.** 19.90%
- D.** 26.10%

51. A risk manager at a distressed debt hedge fund is comparing the credit risk of the fund's investments of GBP 20 million each in two corporate bonds, Bond F and Bond G. The manager models credit losses on the bonds as discrete random variables and makes the following assumptions about their probability distributions:

- Bond F: There is a 1% chance that it will suffer a loss of at least GBP 10 million, but a 0.8% chance it will suffer a loss of the entire GBP 20 million investment.
- Bond G: There is a 1% chance that it will suffer a loss of at least GBP 10 million, but a 0.9% chance it will suffer a loss of between GBP 10 million and GBP 17.5 million, with a loss of more than GBP 17.5 million not considered possible.

Assuming the loss distributions for the two bonds are the same for all losses below the 99 percentile point, which of the following would the manager be correct to conclude?

- A.** The 99% VaR measure is the most effective risk measure for comparing the relative riskiness of the bonds.
- B.** The 99% VaR measure cannot be used in this case, as losses must be assumed to follow a normal distribution to estimate VaR.
- C.** The 99% VaRs of the bonds differ, which accurately represents the bonds as having different levels of risk.
- D.** The 99% VaRs of the bonds are equal, which fails to reflect the fact that the bonds have different levels of risk.

52. A risk manager at a fixed-income hedge fund wants to hedge the interest rate risk exposure of a portfolio. The manager has determined that a 1-bp increase in the 2-year spot rate would decrease the value of the portfolio by INR 1,600, however a 1-bp increase in the 5-year spot rate would increase the value of the portfolio by INR 3,450. The manager plans to hedge the portfolio's exposure using the following two bonds whose key rate 01s (KR01s) are shown below:

Bond	2-year KR01 (INR)	5-year KR01 (INR)
A	48	5
B	4	65

Which of the following transactions would most effectively hedge against the portfolio's interest rate risk?

- A.** Buy 38 of Bond A and short 56 of Bond B
- B.** Buy 75 of Bond A and short 29 of Bond B
- C.** Short 38 of Bond A and buy 56 of Bond B
- D.** Short 75 of Bond A and buy 29 of Bond B

53. The newly hired CRO of a fast-growing multinational bank is reviewing the bank's stress testing process and VaR-based economic capital framework. The CRO wants to ensure that the bank's framework and methodologies reflect best practices put forward by the Basel Committee. In conducting this review, which of the following statements is most appropriate for the CRO to make?

- A.** Traditional VaR measures can be tested to see how well they would have worked in the past, but stressed VaR cannot be effectively back-tested.
- B.** Typically, a VaR-based calculation for market risk capital estimates losses over a multi-year period, while an enterprise-wide stress test focuses on losses over a shorter time period.
- C.** VaR-based economic capital methods and enterprise-wide stress tests typically look at a similar number of loss scenarios.
- D.** Enterprise-wide stress tests typically require modeling the full probability distribution of losses, while VaR-based economic capital estimates do not.

54. A banking supervisor is preparing a set of recommendations to improve the incentive compensation policies in accordance with best practices regarding risk governance and corporate governance. Which of the following actions is the most appropriate for the supervisor to recommend a bank take to improve its compensation structure?

- A.** Require that trading desk managers develop the risk limits used to determine their team's incentive compensation.
- B.** Set incentive compensation levels for all employees of the bank at the same fixed percentage of base salary.
- C.** Assign multi-year guaranteed bonuses for risk managers but performance-based bonuses for senior executives.
- D.** Increase the percentage of incentive compensation that is paid in the form of company stock.

55. The CFO of a property and casualty insurance company is stress testing the impact of different management decisions on the company's operating results and its operating ratio for the coming year. In one of the tested scenarios, the loss ratio increases by 1% and the company pays an additional 1% of its premiums received as dividends. Which of the following correctly describes the impact of this scenario on the company's operating ratio?

- A.** The operating ratio will decrease by 1%.
- B.** The operating ratio will decrease by 2%.
- C.** The operating ratio will increase by 1%.
- D.** The operating ratio will increase by 2%.

56. A first-year analyst on the commodity futures trading desk of a bank is assessing the desk's current portfolio and its typical trading patterns. The analyst examines why a large majority of the trades made by the desk are cash-settled and reviews the delivery mechanics of cash-settled and physically settled futures contracts. Which of the following observations is the most likely for the analyst to make?

- A.** Regulators prefer that cash settlements take place whenever possible due to the inconvenient delivery process involved in physically settling futures contracts.
- B.** The bank should trade physically delivered commodities using forward contracts since they provide a similar payoff to futures contracts, but unlike futures, they trade on exchanges and are less prone to credit risks.
- C.** It is risky for the bank to take physical delivery of commodities due to uncertainty about the exact underlying asset, delivery location, and delivery timeframe.
- D.** Taking physical delivery of commodities can involve warehousing costs and storage costs, so traders usually prefer using futures for financial purposes and not for consumption.

57. The board of directors of a regional bank is reviewing a proposal for a new line of business of offering financing for electric vehicles. The board is concerned that the bank's current risk appetite and risk limit structure may not fully capture potential decreases in the residual value of the vehicles over time, and reviews the bank's risk governance framework to evaluate its soundness. Which of the following is appropriate for the board to recommend after the evaluation of the risk governance framework?

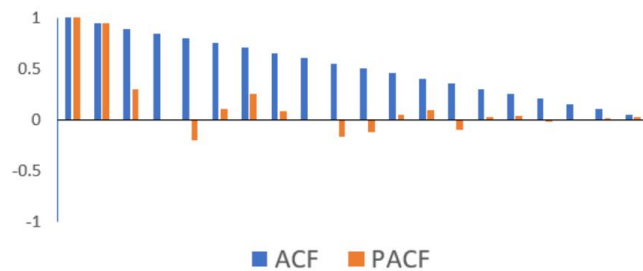
- A.** The bank's risk appetite should be given final approval by senior management.
- B.** The CRO should have the authority to make timely decisions regarding risk limit exceptions for each business line.
- C.** The bank should set Tier 1 risk limits on aggregated risk exposures and Tier 2 risk limits on specific assets or risk types.
- D.** The bank should use delta-normal VaR to set risk limits for counterparty credit risks under stressed market conditions.

58. An analyst working for a large investment firm is evaluating the performance of various investment funds as part of a report to the firm's portfolio managers. One component of the evaluation is the assessment of the statistical moments of the entire population of fund returns. Since the analyst has access to a very large dataset of historical returns, the analyst decides to estimate sample moments such as the sample mean return, sample variance, and sample skewness using a smaller subset of historical returns data. Which of the following would be correct for the analyst to conclude about the sample variance?

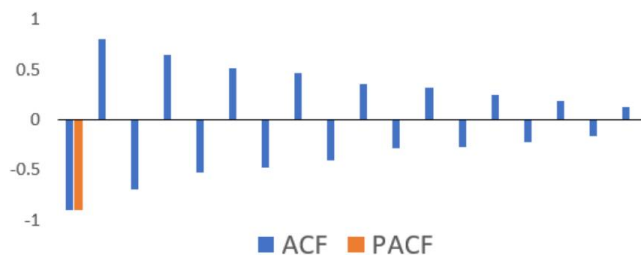
- A.** An unbiased estimator cannot be constructed for the sample variance.
- B.** The sample variance is biased when it is formulated as a function of the population size decreased by 1.
- C.** The sample variance slightly underestimates the population variance.
- D.** The sample variance diverges from the population variance as the size of the sample increases.

59. A quantitative risk analyst at a large financial institution is reviewing the existing model for estimating expected credit loss (ECL) reserves. Upon thoroughly examining the model, the analyst discovers that two key macroeconomic variables, MEV1 & MEV2, need an updated forecast. Before deciding which time series model to apply, the analyst uses statistical software to graph the autocorrelation function (ACF) and partial autocorrelation function (PACF) for each macroeconomic variable and generates the following graphs:

MEV1



MEV2



Based on the graphs above, and supposing that the analyst chose to estimate an AR(1) model, what are the most likely values of the AR parameter (ϕ) in each case?

- A. $\phi < 0$ for MEV1 and $\phi < 0$ for MEV2
- B. $\phi < 0$ for MEV1 and $0 < \phi < 1$ for MEV2
- C. $0 < \phi < 1$ for MEV1 and $\phi < 0$ for MEV2
- D. $0 < \phi < 1$ for MEV1 and $0 < \phi < 1$ for MEV2

60. The board of directors of a manufacturing company has recommended that the company implement a program to hedge some of its steel costs. A risk analyst at the manufacturer is asked to compare the use of options, futures, and forward contracts in establishing the hedges. Which of the following is a correct statement for the analyst to make?

- A. Forward contracts and futures contracts are both typically centrally cleared.
- B. Long options contracts require an up-front payment, while forward contracts do not.
- C. Option contracts are linear derivatives, while forward contracts are non-linear.
- D. Option contracts typically have daily price movement limits set by an exchange, while futures contracts do not.

61. A market risk analyst at a fixed-income hedge fund is assessing the interest rate risk of a portfolio of bonds. The analyst has the following information about the individual portfolio holdings:

Bond	DV01	Convexity	Value (SGD)
1	1,177	108	1,962,000
2	1,265	75	4,216,000
3	2,529	463	6,322,000

Given this information, which of the following is closest to the portfolio's DV01 and convexity?

- A. DV01: 1,657; Convexity: 276**
- B. DV01: 1,657; Convexity: 646**
- C. DV01: 4,971; Convexity: 276**
- D. DV01: 4,971; Convexity: 646**

62. A quantitative analyst on a bank's corporate equity derivatives desk is pricing a 1-year barrier option on a stock. The option has a down-and-in feature specifying that the price in 6 months should be at or below CAD 52.50 for the call option to come into existence. The current price of the stock is CAD 55.00, the strike price of the option is CAD 50.00, the implied volatility of the stock is 12% and the risk-free rate is 1.40%. Using a two-step binomial tree approach, what is the price of the option?

- A. CAD 1.23**
- B. CAD 2.58**
- C. CAD 5.28**
- D. CAD 6.51**

63. A model risk manager is reviewing the output from a prepayment model for a portfolio of recently issued government agency-backed MBS, where balances on underlying mortgages are still relatively high. The model uses a scenario in which interest rates are relatively high and do not change, the equity market declines by 20%, demand for housing does not change, and unemployment increases by 2%. Based upon those inputs, the model indicates that prepayments will increase significantly over the next 6 to 18 months. Assuming the agency backing the MBS remains solvent, which of the following provides the most likely explanation for the increase in prepayments in the model scenario?

- A.** Curtailments are increasing.
- B.** Defaults are increasing.
- C.** More properties are being refinanced.
- D.** Turnover is decreasing.

64. A financial analyst at a large investment firm is evaluating and selecting MBS pools to invest in. The analyst uses a Monte Carlo simulation to value the MBS pools and to determine the option-adjusted spread (OAS) for each of the pools being examined. Which of the following statements regarding OAS and MBS valuation techniques is most appropriate?

- A.** The OAS valuation procedure involves making an initial estimate of the OAS, followed by conducting a Monte Carlo simulation using the yield on US Treasury bills as the discount rate.
- B.** When an MBS pool with a high OAS is identified, it should be purchased immediately to take advantage of this relatively attractive investment compared to other MBS.
- C.** The OAS valuation procedure involves continually changing the OAS estimate until the simulated price is the same as the market price.
- D.** Mortgage rates are perfectly correlated with US Treasury bond rates, and therefore OAS is calculated as the expected MBS return plus the return on US Treasury bonds.

65. A fixed-income trader expects a bull flattening of the interest rate term structure, and wants to pursue a strategy that would profit from this movement in rates. The trader decides to achieve this goal by taking positions in 2-year and 10-year bonds. Will the 2-year rate increase or decrease in the trader's expected scenario, and which of the following sets of trades would be the most likely to generate a profit if the trader's expectations materialize?

- A.** 2-year rate Decrease; Short the 2-year and buy the 10-year
- B.** 2-year rate Decrease; Buy the 2-year and short the 10-year
- C.** 2-year rate Increase; Short the 2-year and buy the 10-year
- D.** 2-year rate Increase; Buy the 2-year and short the 10-year

66. An analyst at a hedge fund is constructing a 95% confidence interval for the 2-month point forecast of the price of a standard option contract on 100 shares of a stock, measured in EUR, using the linear time trend model utilized by the fund. The analyst refers to the model estimated using monthly option prices (OP) over the last 5 years, which is denoted by the following equation:

$$OP_{T+h} = 318.6 + 12.5(T+h) + \varepsilon_{T+h}$$

where T (current month) is equal to 0, h is the horizon of the forecast, and ε_{T+h} is Gaussian white noise. The estimate of the residual standard deviation, σ , is 4.62. What is the correct 95% confidence interval for the price of the option contract?

- A.** [EUR 322.04, EUR 340.15]
- B.** [EUR 334.54, EUR 352.65]
- C.** [EUR 336.02, EUR 351.18]
- D.** [EUR 338.98, EUR 348.22]

67. An intern on the fixed-income trading desk of an investment bank has been asked to assess the relative value of several bonds issued by the same issuer, with the same maturity date, but with different coupon rates. The intern consults a senior trader about appropriate ways to compare the values of the bonds, and the trader explains the implications of the coupon effect. If the term structure of interest rates is upward-sloping and the bonds are correctly priced in the market, which of the following would the trader be correct to state as a result of the coupon effect?

- A.** A bond with a lower coupon rate will be worth more than one with a higher coupon rate.
- B.** A bond with a higher coupon rate will have a lower yield to maturity than one with a lower coupon rate.
- C.** A bond with a higher coupon rate will have a higher Macaulay duration than one with a lower coupon rate.
- D.** A bond with a lower coupon rate will have a lower probability of default than one with a higher coupon rate.

68. A risk manager is reviewing the inherent deficiencies that are commonly observed in risk management frameworks and assessing how risk management systems can be compromised by human agency. The manager focuses on cases involving conflicts of interest, which undermine the credibility of a risk management system. Which of the following is the best example of a conflict of interest?

- A.** The CRO of a bank provides oversight over the second line of defense of the bank.
- B.** A model developer is given responsibility for validating the soundness of the model.
- C.** A senior economist at a regulatory agency helps draft a proposal for a market reform and also advises on trading regulation for the government.
- D.** The head of a business unit of a bank petitions to the risk committee for a temporary risk limit increase for that unit.

69. An FX trader wants to enter into a 6-month EUR/USD forward contract and gathers the following information:

- The spot bid is USD 1.2200 per EUR 1
- 1-month forward ask is USD 1.2208 per EUR 1
- Points for 1-month, 3-month, 6-month and 1-year maturities are 6.0, 22.5, 45.0 and 97.7, respectively
- Points are the same for bid and ask across all maturities

How much will it cost to enter into a long 6-month forward contract?

- A.** USD 1.2155 per EUR 1
- B.** USD 1.2245 per EUR 1
- C.** USD 1.2247 per EUR 1
- D.** USD 1.2253 per EUR 1

70. A risk manager at a US-based fixed-income fund expects a significant decrease in mortgage prepayments. The manager evaluates the impact of this forecast on different types of government and agency securities held in the portfolio and recommends selling the asset type with the greatest downside risk. Assuming all other factors are held constant, and the level of market interest rates is unchanged, which of the following positions should the manager recommend liquidating?

- A.** Interest-only securities (IOs)
- B.** Principal-only securities (POs)
- C.** Pass-through MBS
- D.** US Treasury bills

71. A team of data scientists at a global asset management firm is working on a neural network (NN) model to assess how the choice of metrics used by the firm to select equity investments relates to stock performance. In particular, the team designs the NN model to find the combination of weights for these metrics that generates the highest risk-adjusted return in the firm's portfolios. The team optimizes the model during the training process by varying multiple parameters and observes that, during multiple runs with identical parameters, the loss function converges to different, yet still stable, values. Which of the following is the most appropriate action for the team to take to address this issue?

- A.** Construct a confusion matrix to train the model.
- B.** Apply adaptive boosting when training the model.
- C.** Decrease the learning rate of the model.
- D.** Increase the learning rate of the model.

72. A trader at an agricultural commodity trading company is evaluating the risk of a potential continuous downward trend in orange juice prices. The trader notes that the trading company has entered into a forward contract to purchase a large volume of orange juice 3 months from now. Which of the following actions is most appropriate for the trader to recommend to hedge the risk of a potential decrease in orange juice prices?

- A.** Request a termination of the forward contract on orange juice prior to expiry.
- B.** Purchase a put option on the 3-month orange juice futures contract.
- C.** Enter into another forward contract on orange juice at a lower settlement price.
- D.** Enter into a floating-for-fixed swap on orange juice as the fixed-price payer.

73. A data scientist is presenting to a group of quantitative analysts about the benefits of using machine learning (ML) in developing trading strategies and constructing portfolios. The data scientist compares ML techniques to the classical statistics and econometrics currently employed by the analysts when creating forecasting models. Which of the following correctly differentiates ML from traditional econometric and statistical techniques?

- A.** ML assumes that the data-generating process can be approximated based on economic theory, but traditional statistical techniques do not make this assumption.
- B.** ML does not test a specific hypothesis but lets the data decide on the best features to be included in the model; traditional techniques decide on both the model and the variables.
- C.** Unlike in ML, the role of computer algorithms in traditional econometric and statistical techniques generally goes beyond estimating the parameters and checking for statistical significance.
- D.** Analyses of goodness of fit and residual diagnostics are both relevant for ML but not for the traditional techniques, as they are mainly concerned with the accuracy of forecasts.

74. An analyst on the US Treasury derivatives desk at an investment bank is researching how US Treasury bond futures prices are determined. The analyst examines the assumptions and steps required to calculate the quoted futures price from the prices of Treasury bonds that can be delivered into the futures contract. Which of the following statements is correct regarding this calculation process?

- A.** The quoted futures price can be calculated as the value of a futures contract on an asset providing known income, and the only information necessary to know is the underlying bond's delivery time.
- B.** The price of the bond used in calculating the quoted futures price is the quoted clean cash price minus accrued interest.
- C.** The clean futures price obtained from the clean cash price must be compounded forward at the risk-free interest rate to the time of delivery.
- D.** The quoted futures price is found by dividing the clean futures price by the conversion factor.

75. An analyst at a logistics company is asked to recommend an appropriate hedging strategy using heating oil futures to hedge the volume of jet fuel required by its cargo planes. The analyst gathers the following relevant information about the spot price of jet fuel and the price of the appropriate heating oil futures contract:

- Variance of the change in spot jet fuel prices: 0.0016
- Variance of the change in heating oil futures prices: 0.0009
- Correlation between change in spot jet fuel and heating oil futures prices: 0.8

What is the analyst's best estimate for the optimal hedge ratio?

- A.** 0.60
- B.** 1.07
- C.** 1.33
- D.** 1.42

76. An analyst at an investment bank is researching various hedging techniques used by clients of the bank. The analyst subsequently finds that several clients utilize stack and roll hedging strategies. Which of the following describes the most appropriate motivation for using a stack-and-roll hedging strategy?

- A.** The timing and amount of the risk exposure are not known.
- B.** Cash flows from stack-and-roll hedges are not exchanged until the last futures contract matures.
- C.** The ability to enter short-term hedges at more advantageous prices improves the performance over that of a longer-term hedge program.
- D.** There is greater liquidity in some futures contract maturities than others.

77. A bond portfolio manager purchased a US Treasury bond with a settlement date of September 25. The previous coupon payment was made on May 1, and the next coupon payment will be made on November 1. The manager applies the actual/actual day count convention and determines that there are 184 days in the coupon period, and 37 days between the settlement date and the next coupon payment date. Additional information about the bond is provided below:

- Face value: USD 100
- Annual coupon rate: 3%
- Dirty (cash) price of the bond on settlement date: USD 98.13

What is the clean (quoted) price of the bond on the settlement date of September 25?

- A.** USD 95.73
- B.** USD 96.93
- C.** USD 97.83
- D.** USD 99.33

78. The risk management committee at a US-based bank has appointed a risk consultant to help the committee implement an ERM program. The consultant provides advice on how to set up an ERM framework that is consistent with industry best practices. Which of the following recommendations should the consultant make?

- A.** The bank should separate the decision-making processes of managing credit risk from those of managing market risk to ensure the proper management of its exposures to each risk.
- B.** The bank should develop strategies used to manage enterprise-scale risks at the firm-wide level and implement them across the business lines.
- C.** The bank should build a robust enterprise risk culture that is independent of the views of the board of directors on the value of the bank's business activities in order to avoid any conflict of interest.
- D.** The bank should discourage the integration of the risk management process with the balance sheet management process.

79. A financial risk analyst at a real estate firm is preparing a report on housing prices. The analyst compiles the summary statistics for a dataset with 25 observations and decides to use bias-corrected estimators of the parameters due to the sample size. Which of the following correctly describes the bias of an estimator?

- A.** The bias is typically calculated by taking the square of the deviation between two mean estimators.
- B.** The bias is greatest when the expected value of the estimator is equal to the true population value being estimated.
- C.** The bias is equal to the expected value of the estimator divided by the population value being estimated.
- D.** The bias is the difference between the expected value of an estimator and the true value of the population parameter.

80. An operational risk manager at a bank is preparing for the bank's adoption of the standardized measurement approach (SMA) for calculating operational risk capital according to Basel guidelines. The manager focuses on the formulation of the business indicator component and the loss component. Which of the following statements is correct regarding these two components of the SMA?

- A.** The loss component is calculated as the 99.9 percentile of the 1-year loss for each business unit and risk type.
- B.** The business indicator component is calculated in the same manner under the SMA as it is under the advanced measurement approach.
- C.** Any trading losses suffered by the bank will increase the business indicator component, despite decreasing gross income.
- D.** The SMA is structured in such a way that for an average bank, the loss component is greater than the business indicator component.

81. An investment firm invests in a wide range of asset classes, including private companies, distressed assets, and real estate. A risk analyst at the firm is working on a financial model for evaluating the joint performance of asset classes under extreme market conditions and consults with a risk manager on the project. The manager proposes that the analyst review the concept of cokurtosis and its use in studying how pairs of random variables behave together. Furthermore, the manager advises the analyst to research the three configurations of the cokurtosis measure, (1,3), (2,2), and (3,1), and their corresponding applications. From the discussion, the analyst gathers returns data on two asset classes and decides to calculate the cokurtosis for the two asset returns using the symmetric (2,2) cokurtosis measure. Which of the following would be correct for the analyst to find?

- A.** The symmetric (2,2) measure captures the strength of the relationship between the volatility of one asset class's returns and the volatility of the other asset class's returns.
- B.** The symmetric (2,2) measure captures the likelihood of one asset class's returns taking a significant directional shift whenever the other asset class's returns are large in magnitude.
- C.** If the symmetric (2,2) measure is a large negative value, then the returns on one asset class tend to have the same sign as the returns with extreme values on the other asset class.
- D.** If both asset classes tend to exhibit returns of small magnitude at the same time, then the symmetric (2,2) measure will be large.

82. An asset management firm is planning to launch an open-end mutual fund. Risk managers at the firm are concerned about the potential impact of undesirable trading behaviors on the fund. Which of the following correctly identifies such an undesirable behavior and a potential negative impact of this behavior?

- A.** Accepting large buy orders during market trading hours results in unfair treatment of other shareholders of the fund.
- B.** Allowing frequent buying and selling by large investors could increase costs for all of the shareholders of the fund.
- C.** Prohibiting fund managers from making personal trades in stocks held by the fund could reduce the fund's returns.
- D.** Directing portfolio trades to a brokerage firm that does not offer the mutual fund's shares to its clients could increase reputational risk.

83. A newly hired analyst at an investment bank is using an autoregressive moving average (ARMA) process to model equity returns data. In addition to conducting a visual inspection to assess the model fit, the analyst decides to validate the model by testing for autocorrelation and considers using either the Ljung-Box test statistic or the Box-Pierce test statistic. Which of the following correctly describes these two test statistics?

- A.** The Box-Pierce test statistic is equal to the sum of the squared autocorrelations scaled by the sample size.
- B.** The Box-Pierce test statistic is the sum of autocorrelations that follow a converging Chi-square distribution.
- C.** Unlike the Box-Pierce test statistic, the Ljung-Box test statistic follows an asymptotic standard normal distribution.
- D.** Unlike the Box-Pierce test, the Ljung-Box test does not assume that the estimated model residuals follow a white noise process.

84. A risk manager at a transportation logistics company is evaluating an existing hedge. The hedge is designed to protect the company against changes in the price of jet fuel by using heating oil futures, and the manager is checking if the hedge employs the optimal number of contracts. The manager collects the following information about the relevant current prices and those prevailing when the hedge was established:

Amount of jet fuel to be hedged	1,500,000 gallons
Price of jet fuel when hedge was established	USD 1.25 per gallon
Amount underlying each heating oil future	42,000 gallons
Price of heating oil futures when hedge was established	USD 1.30 per gallon
Correlation between the price of jet fuel and price of heating oil futures contract	0.85
Standard deviation of daily returns of price of jet fuel	0.03
Standard deviation of the change in the price of jet fuel over the life of the hedge	0.03
Standard deviation of daily returns of price of heating oil futures	0.04
Standard deviation of the change in the price of heating oil futures over the life of the hedge	0.04
Current price of jet fuel	USD 1.00
Current price of heating oil futures	USD 1.15

The manager notes that an optimal hedge was initially established and now wants to apply a tailing-the-hedge adjustment. After the adjustment is made, what is the correct hedge position for the company to have in place that now reflects both the current price of jet fuel and heating oil futures?

- A.** Long 20 contracts
- B.** Long 23 contracts
- C.** Short 20 contracts
- D.** Short 23 contracts

85. The treasurer of a large manufacturing company has decided to hedge against rising interest rates. The treasurer wants to enter into a 2-year fixed-for-floating swap with a notional of USD 100 million, a fixed annual interest rate of 2.75%, semi-annual payments, and a floating interest rate of 6-month SOFR plus 30 bps, starting in January of Year 1. The treasurer uses the following forecast of future 6-month SOFR rates:

Time period	Rate
Jan-Jun Year 1	2.40%
July-Dec Year 1	2.57%
Jan-Jun Year 2	2.66%
July-Dec Year 2	2.70%

Which of the following is the best estimate of the net cash flow that the treasurer expects for the company to receive at the end of year 2?

- A. USD –125,000
- B. USD –25,000
- C. USD 125,000
- D. USD 265,000

86. A newly hired junior analyst at a US-based bank is researching reference interest rates currently or historically used in various types of financial markets. The analyst studies how different interest rates are determined, and how they are applied to various debt instruments. Which of the following statements correctly describes the specified interest rate?

- A. Overnight interbank borrowing rates are the overnight swap rates that the US Federal Reserve sets for the banks to charge each other to lend cash overnight.
- B. A repo rate is the interest rate charged by one party to another on unsecured overnight borrowings.
- C. An interest rate on government debt issued by any country in any currency can be considered risk-free and used for discounting purposes.
- D. The Secured Overnight Funding Rate (SOFR) is based on the repo rates in actual repurchase agreement transactions, while LIBOR was based on estimated borrowing rates.

87. An investment manager at a venture capital (VC) firm is evaluating a decision to invest in two promising start-up companies focusing on fintech applications. The manager analyzes the following information:

- For each company, there is a 30% probability of successfully launching a usable fintech application.
- The success of one company in launching a usable fintech application is independent of the success of the other.
- If either of the two companies is successful, the VC firm will generate USD 40 million in profit.
- If both companies are successful, the VC firm will generate USD 70 million in profit.
- If neither company is successful, the VC firm will incur a loss of USD 15 million.

What is the expected value of the VC firm's profit from investing in the two companies?

- A.** USD 6.30 million
- B.** USD 7.35 million
- C.** USD 15.75 million
- D.** USD 16.80 million

88. A derivatives dealer actively trades options on various underlying assets with its clients. The firm wants to apply the Black-Scholes-Merton (BSM) model to price a call option on a futures contract. Relevant data is provided below:

- Current futures price: EUR 63
- Strike price of the option: EUR 68
- Time to expiration of the option: 6 months
- Time to maturity of the underlying futures contract: 18 months
- Continuously compounded annual risk-free interest rate: 3%
- $N(d_1)$: 0.4678
- $N(d_2)$: 0.3449

Which of the following is closest to the value of this option estimated using the BSM model?

- A.** EUR 5.75
- B.** EUR 5.93
- C.** EUR 6.36
- D.** EUR 6.81

89. A junior fixed-income analyst at an investment management firm is evaluating the credit quality of a diversified bond portfolio consisting of bonds classified as investment grade, high-yield, or unrated. Upon observing that the expected return profile of a bond appears to be related to its classification, the analyst constructs a probability matrix to assign probabilities to different combinations of a bond's return profile and its classification. Which of the following correctly describes this probability matrix?

- A.** The probability matrix is a multivariate representation of a cumulative density function (CDF) which relates the outcomes of a bond's return profile and its classification.
- B.** The probability matrix is a tabular representation of the probability mass function (PMF) which relates the outcomes of a bond's return profile and its classification.
- C.** Each cell in the probability matrix contains the probability that a combination of three outcomes is realized.
- D.** The marginal distribution of a random variable in the probability matrix can be computed by multiplying the probability of the three classification outcomes.

90. A risk manager at a bank is explaining various types of assets to a junior analyst. The risk manager discusses the characteristics of financial, non-financial, investment, and consumption assets and gives examples of each. Which of the following is the best example of a consumption asset?

- A.** A stockpile of copper owned by an electrical component manufacturing company
- B.** A single-family residential rental property that generates a 5.5% annual return in rental income
- C.** A position in gold ETFs that a municipality periodically partially liquidates to fund its ongoing business operations
- D.** The land and a manufacturing facility owned by a company that uses them for its primary business

91. A fixed-income portfolio manager is using discount factors to price a sovereign bond. The bond is a coupon bond with the following cash flows:

- 3 months from today: USD 67,500
- 6 months from today: USD 67,500
- 9 months from today: USD 4,567,500

The manager observes the following market price quotes for three zero-coupon bonds issued by the same country:

Time to maturity (months)	Face value (USD)	Market price (USD)
3	100	97.8012
6	100	95.2375
9	100	92.3805

Based on the discount factors of the zero-coupon bonds, what is the present value of the cash flows from the sovereign coupon bond?

- A.** USD 3,728,209
- B.** USD 4,059,055
- C.** USD 4,349,780
- D.** USD 4,436,915

92. An equity analyst on a trading desk at an investment bank is comparing different investment vehicles. The analyst studies the characteristics of open-end mutual funds, closed-end mutual funds, and ETFs. Which of the following is a correct comparison of these investment vehicles?

- A.** The number of shares in open-end and closed-end mutual funds remains constant over time, while the number of shares in an ETF fluctuates.
- B.** Trades in open-end and closed-end mutual funds are executed once a day at the net asset value, while ETFs can be traded at the market price any time the market is open.
- C.** Open-end mutual funds need to hold some liquid assets to meet redemptions, while closed-end mutual funds and ETFs do not need to hold any liquid assets on hand.
- D.** Open-end and closed-end mutual funds typically trade at or very close to their net asset values, while ETFs typically trade at prices lower than their net asset value.

93. A quantitative analyst supporting the acquisitions team of a European corporate real estate firm is using the decision tree technique to create a model for forecasting property prices. The analyst compiles a training data set comprised of information from 10 recent property sales, as shown in the following table:

Property	Use of site	Occupancy status (Y=occupied)	Expected positive cash flow	Sale price greater than EUR 8,000,000
1	Office	Y	Y	Y
2	Retail	N	Y	N
3	Retail	Y	N	N
4	Office	N	Y	Y
5	Retail	N	Y	Y
6	Retail	N	N	N
7	Office	N	Y	N
8	Retail	Y	Y	Y
9	Retail	N	N	N
10	Retail	Y	Y	Y

The table also includes the target variable of the model: a class label indicating whether the property was sold for a price greater than EUR 8,000,000. The analyst selects the occupancy status as the feature that is used as the root node of the decision tree. What is the estimated information gain of the split put forward by this root node?

- A. 0.08
- B. 0.37
- C. 0.44
- D. 0.82

94. An equity research analyst specializing in the consumer goods sector is modeling retail sales data that exhibits seasonality using an autoregressive moving average (ARMA) process. The analyst notes that the ARMA process used in the model is covariance stationary. Which of the following correctly describes how seasonality is modeled in a covariance-stationary ARMA process?

- A. Seasonality can be introduced to an ARMA process through the autoregressive (AR) component, but not through the moving average (MA) component.
- B. Seasonality can be introduced to an ARMA process through the moving average (MA) component, but not through the autoregressive (AR) component.
- C. Seasonality is included in an ARMA model by multiplying the number of full seasonal cycles by the lag lengths of the ARMA process.
- D. Seasonality is included in an ARMA model by multiplying the short-run lag polynomial by a seasonal lag polynomial.

95. A credit risk analyst is running scenarios on the impact of correlation on the standard deviation of percentage loss for middle market loan portfolios. The analyst is using a binomial distribution to analyze a portfolio of 7 loans. The principal amount of each loan is USD 125,000. The analyst assumes that each loan has a recovery rate of 55%, a default probability of 6%, and a default correlation of 0.35 with all other loans in the portfolio. Which of the following is closest to the ratio of the portfolio standard deviation of losses to the size of the portfolio position?

- A.** 0.07
- B.** 0.11
- C.** 0.19
- D.** 0.23

96. A risk analyst at a derivatives trading firm is estimating the VaR of several options positions. The analyst considers using the delta-normal model for these estimates, but acknowledges this model has limitations that could make its application less suitable under certain situations. The analyst focuses on the following positions in options on stock LCO:

Option	Type	Strike price (USD)
A	Call	75
B	Call	80
C	Put	80
D	Put	95

The current price of stock LCO is USD 79, and all four options expire in exactly 1 month. For which of these positions would delta-normal VaR best reflect its risk?

- A.** Option A
- B.** Option B
- C.** Option C
- D.** Option D

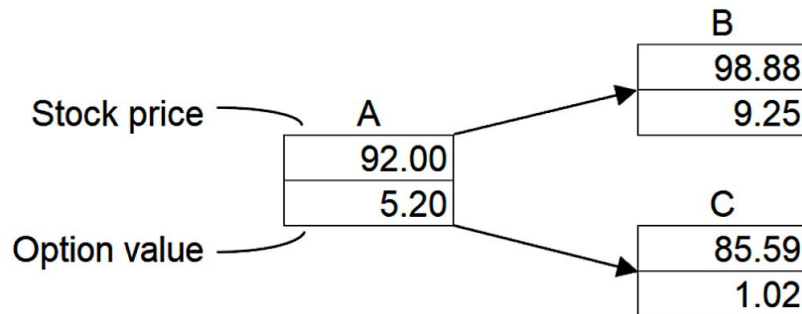
97. An analyst at an investment company is researching the characteristics of American-style options. The analyst focuses on the ability and motivations of American-style option holders to exercise the option prior to maturity. Which of the following statements is correct regarding the early exercise of American-style stock options?

- A.** Exercising an American-style call option early allows the holder to capture the time value of the option as well as its intrinsic value.
- B.** An American-style put option on stocks that pay no dividends should never be exercised early, since the option holder would give up the opportunity to profit from a drop in the stock price.
- C.** An increase in dividends makes it more attractive for the holder of an American-style call option to exercise the option early.
- D.** A decline in interest rates makes it more attractive for the holder of an American-style put option on a stock paying dividends to exercise the option early.

98. A certified FRM who is a risk manager at a multinational bank has been asked to explain the GARP Code of Conduct to several junior risk analysts. The manager uses several potential work-related scenarios to illustrate the Code of Conduct. Which of the following scenarios exemplifies behavior that complies with the standards of the GARP Code of Conduct?

- A.** A wealth manager at the bank's asset management unit uses information from an investment analysis that utilized proprietary information to enter into a profitable trade on behalf of a client.
- B.** A compliance officer working in a foreign office of the bank declines a request to take an action that is considered ethical in the bank's home country but not in the officer's local country.
- C.** A model validator at the bank relies on the opinions of the quantitative analyst who developed the model to evaluate the soundness of the model during the validation process.
- D.** A risk analyst at a bank analyzes a trade and concludes that losses will not exceed the trade's 10-day, 95% VaR limit; therefore, there is no risk of losses exceeding the VaR.

99. A junior trader at an equity hedge fund is assessing the market risk of an option position. The option is a call option on stock MTP, which is currently trading at USD 92 per share. The trader has valued the option using a binomial tree, the three left-most nodes of which are shown below:



Given this information, and assuming the length of each time step in the tree is very short, which of the following conclusions would the trader be correct to reach?

- A. The option delta is approximately 0.62, therefore a portfolio consisting of a short position in 62 options and a long position in 100 shares would have the same value at node B or node C.
- B. The option delta is approximately 0.62, therefore if the stock price increases by USD 1, the option price would increase by approximately USD 0.62.
- C. The option delta is approximately 0.94, therefore a portfolio consisting of a short position in 94 options and a long position in 100 shares would have the same value at node B or node C.
- D. The option delta is approximately 0.94, therefore if the stock price increases by USD 1, the option price would increase by approximately USD 0.94.

100. A trader on the options trading desk at an investment bank is discussing the Black-Scholes-Merton model with an intern. The trader mentions that while various assumptions were used to originally derive the model, several of those assumptions have been subsequently relaxed to allow for the pricing of options under a wider array of circumstances. However, the trader stresses that some other assumptions have not been relaxed and are still considered necessary. Which of the following assumptions is necessary for the Black-Scholes-Merton model to be used in pricing options?

- A. The underlying asset does not pay any dividends during the life of the option.
- B. The option under consideration cannot be an American option.
- C. The execution of risk-free arbitrage opportunities is not possible.
- D. The underlying asset's price at the end of a 1-year period will be normally distributed.